सत्यमेव जयते

GOVERNMENT OF MAHARASHTRA

A REPORT ON FLOODS 2019 (KRISHNA SUB-BASIN)

by

EXPERTS STUDY COMMITTEE

(Analysis,Causes and Remedies)

Volume- I
Main Report

Volume- II
Annexures

Volume- III
Maps

May 2020

EXPERTS STUDY COMMITTEE: FLOOD 2019

KRISHNA SUB-BASIN

BHIMA SUB-BASIN

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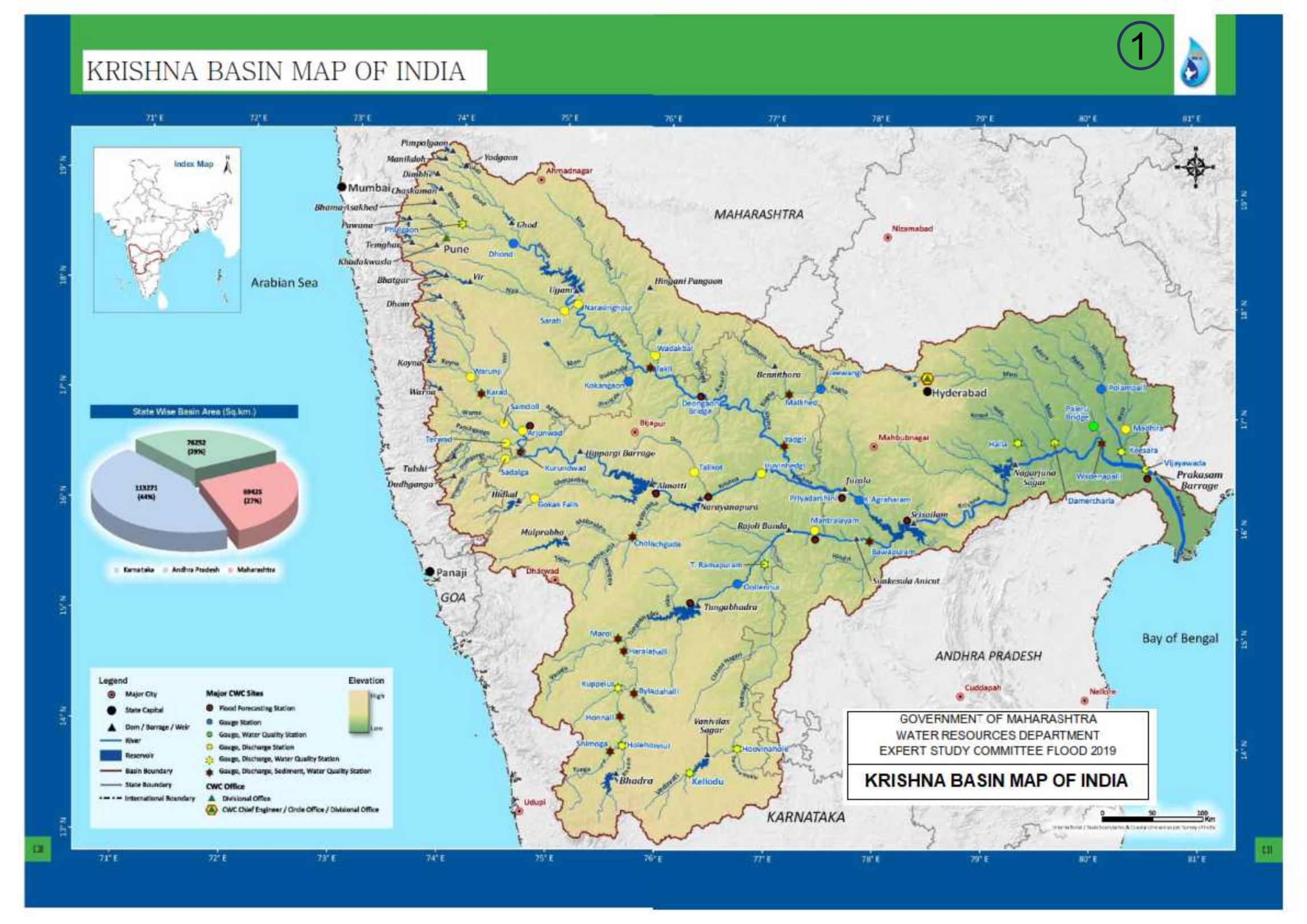
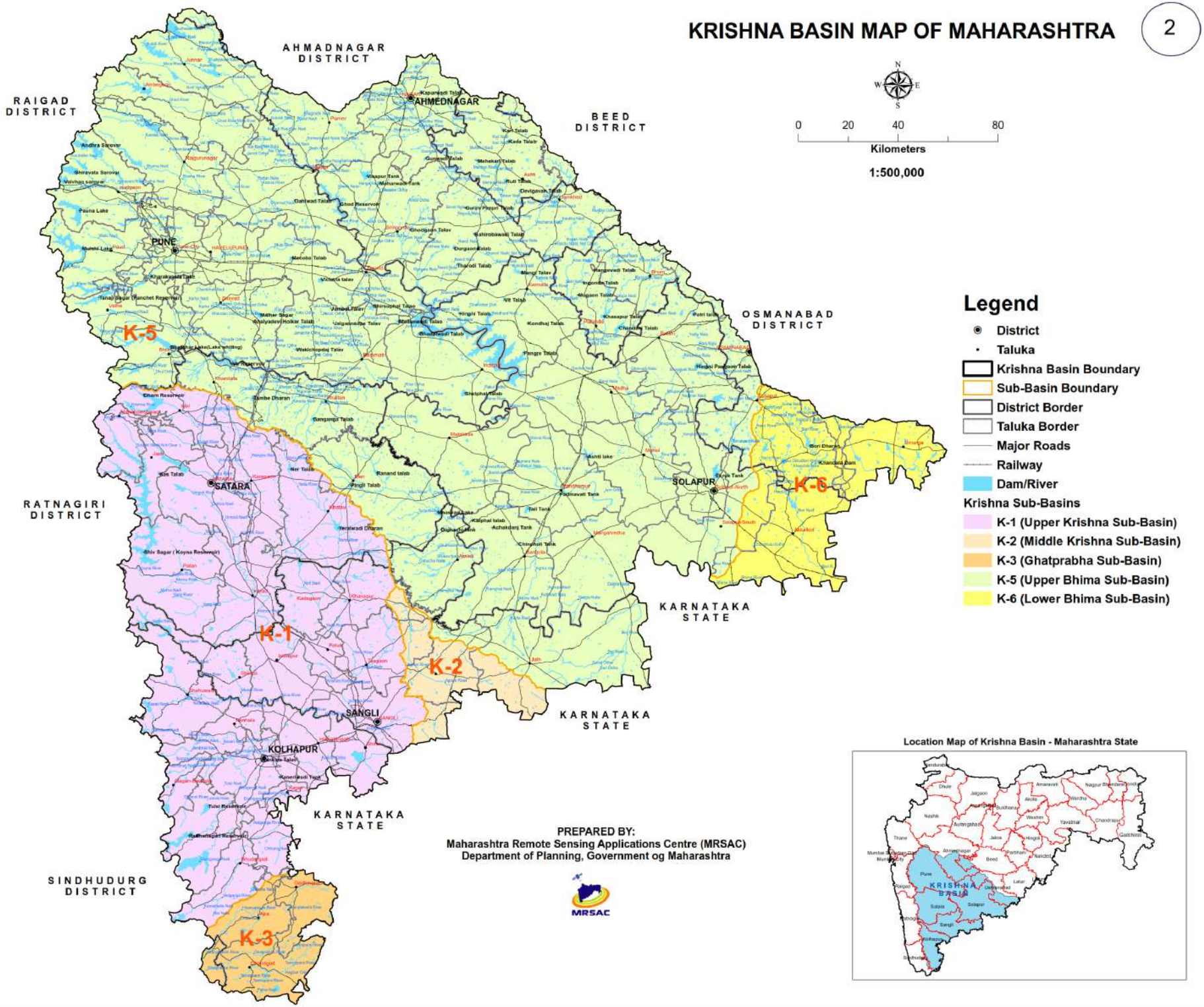
KRISHNA BASIN MAP OF INDIA



KRISHNA BASIN MAP OF MAHARASHTRA



Upper Krishna Sub-Basin (K-1)

MAJOR DRAINAGE MAP

N
W E
S

0 12.5 25 50

Kilometers

1:300,000

Dhom Reservoir
Krishna River
Chandrabhaga Nala
Koyna River
Krishna River
Vena River
Vasna Nadi
Kudali River
Solayhi Nadi(Not Clear)
Wangna Nadi
Yerla River
Krishna River
Ner Talab
Kas Talab
Vena River
Vasna Nadi
Yerla River
Krishna River
Urmodi River
Dolhira Nadi
Urmodi Nadi
Yeralwadi Dharan
Tarli Nadi
Krishna River
Yerla River
Kera Nadi
Shiv Sagar (Koyna Reservoir)
Chand Nadi
Koyna River
Koyna River
Morna River
Nini Nadi
Yerla River
Morna Nadi
Vang River
Vang Nadi
Krishna River
Varna Nadi
Yerla River
Varna Nadi
Yerla River
Kansa Nadi
Kapur Nala
Varna River
Krishna River
Yerla River
Morna River
Kadvi Nadi
Varna River
Shali Nadi
Kadri Nadi
Krishna River
Kasari Nadi
Ambarde Nadi
Kandra Nala
Varna River
Varna River
Varna River
Krishna River
Garnavati Nadi
Kasari Nadi
Kasari Nadi
Kasari Nadi
Kasari nadi
Jambhali Nadi
Panchganga River
Panchganga River
Kumthi Nadi
Kabnur Odha
Saraswati Nadi
Dhamni River
Rankala Talav
Jambhali Nadi
Kumbhi River
Kaneriwadi Tank
Tulsi Nadi
Bhogavati Nadi
Dudhganga River
Dhamni River
Lonsla Nala
Dudhganga River
Tulsi Reservoir
Vedganga River
Radhanagari Reservoir
Chikutra Nadi
Nganga Nadi
Chikutta Nadi
Vedganga River
Kamathalla Nala
Mor Nala

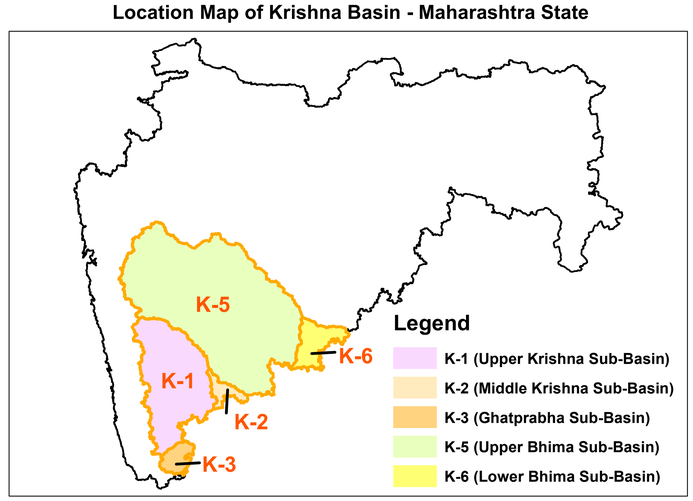
Legend

- Major Drainage and Reservoir
- Krishna Basin Boundary
- Sub-Basin Boundary

PREPARED BY:
Maharashtra Remote Sensing Applications Centre (MRSAC)
Department of Planning, Government of Maharashtra

MRSAC

Location Map of Krishna Basin - Maharashtra State



MAP SHOWING KRISHNA RIVER UPTO ALMATTI DAM

NORTH

SCALE 1:32500

SATARA DISTRICT

SANGLI DISTRICT

KOLHAPUR DISTRICT

RATNAGIRI DISTRICT

BAGALKOT DISTRICT

KARNATAKA STATE

MAHARASHTRA STATE

KRISHNA RIVER

VENNA RIVER

URMODI RIVER

TARLI RIVER

KOYANA RIVER

YERLA RIVER

MANDHAL RIVER

MORNA RIVER

WARNA RIVER

PANCHGANGA RIVER

DUDHGANGA RIVER

VEDGANGA RIVER

HIRANYAKESHI RIVER

GHATPRABHA RIVER

ARKAVATI RIVER

DHAMANI RIVER

KASARI RIVER

KUMBHI RIVER

SALVAN RIVER

TILARI RIVER

JAMBHALI RIVER

DHAMNI RIVER

DHAMANA RIVER

VASANA PROJECT

NER PROJECT

YERALWADI PROJECT

UPALE WANGI PROJECT

URMODI PROJECT

TARLI PROJECT

KERA PROJECT

WANG PROJECT

YEOTI MASOLI PROJECT

MORNA PROJECT

KADVI PROJECT

KUMBHI PROJECT

DHAMNI PROJECT

RADHANAGARI PROJECT

DUDHGANGA PROJECT

PATGAON PROJECT

CHIKOTRA PROJECT

BASAPPAWADI PROJECT

SIDHEWADI PROJECT

DHOM DAM

KANHER DAM

KOYANA DAM

WARNA DAM

Almatti Dam

Sangam UKP Dam

Kollar Weir

Galagali Weir

Hipparagi Dam

Hidkal Dam

Mhaisal Weir

Rajapur Weir

Kallol Weir

IRWIN BRIDGE

SANGLI BRIDGE

BHILWADI BRIDGE

TAKARI BRIDGE

BAHE BRIDGE

KANVALI BRIDGE

KHIDRAPUR WEIR

NIMANI WEIR

10.72 K.M.

8.0 K.M.

2.0 K.M.

2.86 K.M.

6.42 K.M.

5.00 K.M.

SATARA

KARAD

PATAN

PALUS

TASGAON

KADEGAON

WALWA

MIRAJ

SANGLI

SHIROL

RAIBAG

ATHANI

JAMAKHANDI

BILAGI

MUDHOL

BAGALKOT

TAKARI

KHANAPUR

VITA

KHANAPUR

HATKANANGALE

KOLHAPUR

KAGAL

KRAVIR

SHAHUWADI

PANHALA

GAGANBAWADA

BHUDARGAD

GADHINGLAJ

AJARA

GOKAK

BELGAUM

CHIKODI

RADHANAGARI

KARANJEN

KADAVE

KARVE

TAMBVE

NAVDI

TRIPUTI

JAMDADVADI

MANDRUL

MOREWASTI

NISARE

KASEGAON

SHIRALE

KHARI WADI

NARSHAPUR

DHOTREWADI

YEWALEWADI

BEDAG

MACHCHHINDRA

BERADMACHI

KASEGAON

KOLE

WATHAR

KALWADE

MALKHED

DUSHERE

SHERE

RETHRE KH.

RETHRE BK.

ATAKE

BAHE

TAMBVE

KOLE

KUSUR

TARUKH

TULSAN

UNDALE

MANU

KASARSIRAMBE

ONDOSHI

ONDE

GUDHE

LATKEWADI

TALEGAON

SHELEWADI

MALKAN

SONWADE

DHAMANT

UTHANE

TOLEWADI

RISWAD

KHUDALAPUR

CHANDOLI BK.

NANDOLI

MANADUR

AGALE

BATEWADI

MARATHWADI

KHUJGAON

KALKUNDRE

SONDOLI

KHEDE

CHINCHOLI

MALEWADI

KOKARE

NERLE

BHEDSGAON

KOLOLI

AKURLE

MANGAON

WADGAON

SAVE

SHIVARE

GOGAVE

WADICHARAN

PUSARLE

ALTUR

KARUNGALE

MALKAPUR

CHANWAD

KASARDE

PARALE

SUTARWADI

MAN

AINWADI

BHOGAON

GHUNGUR

NANUNDRE

TELVE

PUNAL

KODOLE

WAKANI

WASHI

DEVALE

BEED

MHANU

ASARE

KOTHALI

CHUE

HALADI

EKONDI

BHAMATE

KALE

MALHARPETH

PARKHANDALE

SAVARDE

GOVE

UPWADE

SADOLI

MHALSWADE

AMBARDE

PANORE

PARITE

KURUKALI

TISANGI

NIVADE

VETAVADE

SALVAN

OHONDE

YELWADE

KAUDAV

CHANDOLI

TILVE

AVALI BK.

ANAJE

GUDAL

AKNUR

GUDALWADI

KEOLOSHI

MHASURLI

DHAMANE

DHUNDWADE

KHOKURLE

MANDUKALI

PADALWADI

ANDUR

BARKI

INJOLI

KARANJEN

TANDULWADI

SAVARDE

KHANAPUR

BANKRE

KARANJIVANE

NIDHORI

GHOTWADI

KASBA WALWA

SONALI

MALMADE

WAGHAPUR

NADYAL

KASARI

HASUR

BORWADE

KOOR

GANGAPUR

MASA

MADILAGE

CHIMGAON

HANBARWADI

KANDGAON

MADWAL

PANGERI

PIMPALGAON

UNTUR

KOUDGE

MANWAD

KILLE

ANALGUNDI

TIKKEWADI

KONWADE

AINI

HEDWADE

MINCHE BK

PHYE

YERANDP

NISHNAP

KONOLI

WAKI

ADOLI

KADGAON

VESARDE

KODOSHI

PARPOLI

MARI

SAVARDE

DAJIPUR

RASHIWADE

PADALI

DURGMANWAD

MARLWAD

PHEGIVADE

RAJAPUR

SHENGAON

KALLOLI

ANKALI

SIDOAPURWADI

EKSAMBA

NANDIKURLI

KADAPUR

KERUR

DATTWAD

GHOSARWADI

SHIRADWAD

ABDUL LAT

TAKALWADI

ICHALKARANJI

ICHALKARANJI K. T. WEIR

SHIROL BK.

NURSINHWADI

KUROLI

DHARANGUTTI

RAJAPURWADI

MAJAREWADI (BUVACHE WATHAR)

WATHAR TRF UDYAGAON

TERWAD WEIR

TERWAD

AKLWAT

TAKLI

DANWAD

BUBNAL

KHIDRAPUR

KUMBHOJ

NEJ

TAISGAON

VADGAON KASABA

VATHAR TARF VADGAON

AMBAP

SAMBHAPUR

TOP

CHOKAK

MANGAON

PATTAN KODOLI

WASAGADE

RENDAL

HUPARI

INGALI

RUI CHANDUR WEIR

RUI

SAJANI

TIWARI

KOROCHI

ATGIRE

NANDANI

HAROLI

JAMBHALI

SHIRDHON

TAKAVADE

ABDUL LAT

SHIRADWAD

GHOSARWADI

DATTWAD

MALIKWAD

NANADI

RUPNAR

EKSAMBA

NANDIKURLI

KADAPUR

KERUR

SIDOAPURWADI

NIPANI

DANDAPUR

RAJPUR

DEVANKATTI

DEVAPURHATTI

KANKANWADI

NAGNUR

DHARAMATTI

SUNDHAL

KULLIGOD

HUNSHAL

UDAGATTI

HADAGINAL

KALARKOD

MELLYANKI

CHIKKODI

NALLANHATTI

ADIHATTI

BHENDIWAD

KATARBHAVI

MAVINHUNDI

MIPURHATTI

JODHPATTI

BAJEWADI

BADIGWAD

SANGANKERI

BALLOBAL

PULAGADDI

VADERATTI

MUDALGI

RANGAPUR

KAMALDINNI

BHAIRAWADI

LAXMESWAR

HOSATTI

KULIGOD

TIGADI

PULAGADDI

SUNDHAL

DHARAMATTI

NAGNUR

KANKANWADI

DEVAPURHATTI

DEVANKATTI

KANKIKUR

KABBUR

HANCHINAL

KANCHANHATTI

BELKUD

ARTHAVI

SINDIKURBET

FALLS

GOKAK

SADALGA

NEJ

NAGNAL

HINUKUDI

CHIKODI

GINGAON

CHINCHNI

KANSOLI

UMRANI

GEMBRAWAD

SHIRHATTI

SADALGA

GYDAS

KHANAPUR

COAKRS

MALMATTI

YAMRAHTTI

PARNANATTI

CHINMALKI

KADBGATTI

PANCHAPUR

SULTANPUR

MANGANOOR

KANDRATTI

KOVINATTI

MIRVANIR

ARKITTI

ANKALGI

HONNIHAL

MANNIKERI

TEMPLE

MALLAYYA

GOVINA KOPPA

TOLAMATTI

HERKAL

TUMMARAMATTI

YALLIGUTTI

SIDDAPUR

NAGANUR

BADAGANDI

SUNAG

KADAPATTI

ANAGWADI

TEGGI

BISANAL

JANMATI

BAVLATTI

UDAGATTI

SHADONAL

KALADAGI

GUDDANEKARI

SIMMIKERI

TOLLIGERI

KATARAKI

DEVANAL

ARAKERI

KUNDARAGI

SOKNADAGI

GRDOANAKERI

MARBAL

MALLIGERI

YADAHALLI

AMAJERI

GANIS

BUDNI

KONNUR

MAREGUDDI

MANTUR

HADAGALLI

ALGUND

ANANTPUR

MAGHAKNUD

BANTANUR

KARKALMATTI

KATARAKI

KALADGI

HOUDAPUR

CHINCHKHANDI

HAIKI

KOPP

KMANA

MALLAPUR

MANAKADUR

YARAGUDI

TIMMAPUR

YADWAD

PUR

VENKAT

HUNSHAL

ARALIMATTI

MADABAVI

DHAVALESWAR

SANGANATTI

MIRJI

RANJANAGI

NAGANUR

GULBHI

CHINCHKHANDI

HOUDAPUR

KALADGI

KARKALMATTI

BANTANUR

GUDDANEKARI

SIMMIKERI

TOLLIGERI

KATARAKI

DEVANAL

ARAKERI

KUNDARAGI

SOKNADAGI

YADAHALLI

GRDOANAKERI

MARBAL

MELLIGERI

YADAHALLI

AMAJERI

GANIS

BUDNI

KONNUR

MAREGUDDI

MANTUR

HADAGALLI

ALGUND

ANANTPUR

MAGHAKNUD

BANTANUR

KARKALMATTI

KATARAKI

KALADGI

HOUDAPUR

CHINCHKHANDI

HAIKI

KOPP

KMANA

MALLAPUR

MANAKADUR

YARAGUDI

TIMMAPUR

YADWAD

PUR

VENKAT

HUNSHAL

ARALIMATTI

MADABAVI

DHAVALESWAR

SANGANATTI

MIRJI

RANJANAGI

NAGANUR

GULBHI

Upper Krishna Sub-Basin (K-1)

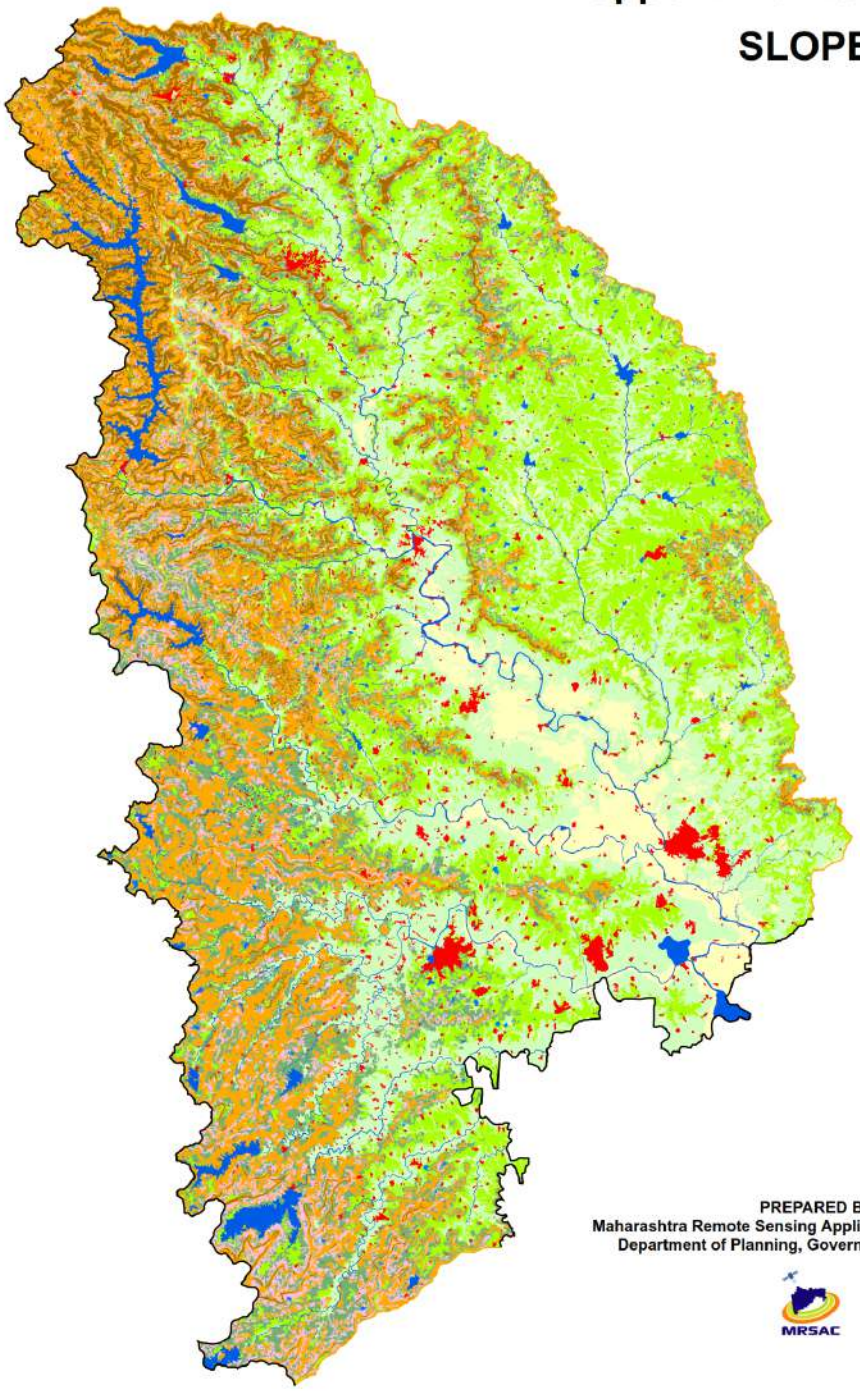
SLOPE MAP

N
W E
S

0 12.5 25 50

Kilometers

1:300,000



Legend

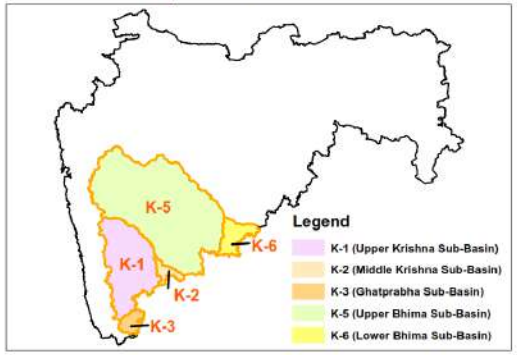
SLOPE

- 0 - 1 %
- 1 - 3 %
- 3 - 5 %
- 5 - 10 %
- 10 - 15 %
- 15 - 35 %
- 35 - 50 %
- Habitation
- Waterbody

PREPARED BY:
Maharashtra Remote Sensing Applications Centre (MRSAC)
Department of Planning, Government of Maharashtra

MRSAC

Location Map of Krishna Basin - Maharashtra State



LINE DIAGRAM OF RIVER SYSTEM IN KRISHNA K-1 SUB-BASIN

LEGEND

KRISHNA MAIN RIVER
MAJOR TRIBUTORIES
OTHER TRIBUTORIES
SUB-TRIBUTORIES
M.W.L. Maximum Water Level
F.R.L. Full Reservoir Level
M.D.D.L. Max.Draw Down Level
SPL.Q. Spillway Capacity

DAM
WITH CONTENT IN MCM
LEVELS IN METER

M.W.L.815.81
F.R.L. 813.50
M.D.D.L 802.84
SPL.Q. 1479.0

11.78
R.B.L. 794.32
NER DAM

M.W.L.700.80
F.R.L. 698.20
M.D.D.L 694.40
SPL.Q. 2083.00

32.80
R.B.L. 683.30
YERALWADI DAM

YERALA RIVER

M.W.L.814.625
F.R.L. 814.000
M.D.D.L 768.544
SPL.Q. 923.14

M.W.L.748.30
F.R.L. 747.70
M.D.D.L 725.40
SPL.Q.1840.34

115.532
382.32
R.B.L. 701.09
DHOM DAM
(495 KM)
DHOM BALAKAWADI DAM

KRISHNA RIVER

UDATARE (467 KM)
MAHULI (444 KM)
KASHIL (402 KM)
UMBRAJ (391 KM)
SHIVADE (389 KM)
KARAD (366 KM)
BRAHMNAL (271.60 KM)
SANGLI (259KM)
KURUNDWAD (220.5 KM) Narsoba Wadi
RAJAPUR WEIR (211 KM)
TOWARDS KARNATAKA

VENNA RIVER
URMODI RIVER
TARALI RIVER
UTTARMAND RIVER
WANG RIVER
WARNA RIVER
PANCHAGANGA RIVER

NAGEWADI DAM
6.47
R.B.L. 814.280

KANHER DAM
286
R.B.L. 651.00

M.W.L.851.300
F.R.L. 848.800
M.D.D.L 829.000
SPL.Q. 325.000

M.W.L.690.98
F.R.L. 690.78
M.D.D.L662.025
SPL.Q. 1777.97

URMODI DAM
282.14
R.B.L. 650.50

M.W.L.696.50
F.R.L. 696.00
M.D.D.L 665.65
SPL.Q. 1940.80

TARALI DAM
165.71
R.B.L. 642.27

M.W.L.711.93
F.R.L. 711.30
M.D.D.L 649.30
SPL.Q. 1721.00

UTTARMAND DAM
24.925
R.B.L. 646.350

M.W.L.688.300
F.R.L. 687.500
M.D.D.L 658.000
SPL.Q. 824.000

KOYNA RIVER
KOYNA DAM
2980.00
R.B.L. 579.12

M.W.L.659.89
F.R.L. 659.434
M.D.D.L 611.124
SPL.Q. 5743.02

MORANA RIVER
MORANA GUREGHAR DAM
39.55
R.B.L. 617.48

M.W.L.662.0
F.R.L. 660.40
M.D.D.L 635.0
SPL.Q. 846.000

WANG DAM
77.29
R.B.L. 608.82

M.W.L.657.33
F.R.L. 655.85
M.D.D.L 617.50
SPL.Q.1722.00

DAKSHINMAND RIVER
YEOTI MASOLI DAM
07.30
R.B.L.

M.W.L.
F.R.L.710.60
M.D.D.L
SPL.Q.330.000

KUDALI RIVER
KUDALI MAHU DAM
37.434
R.B.L. 810.385

M.W.L.862.235
F.R.L. 861.635
M.D.D.L 821.310
SPL.Q. 705.000

KUDALI HATTEGHAR DAM
7.371
R.B.L. 800.645

M.W.L.837.200
F.R.L. 835.200
M.D.D.L 811.080
SPL.Q. 150.000

MORNA SHIRALA DAM
21.24
R.B.L.

M.W.L.
F.R.L.595.70
M.D.D.L
SPL.Q.1341.57

WARNA DAM
974.180
R.B.L. 556.00

M.W.L.628.050
F.R.L. 626.900
M.D.D.L 588.200
SPL.Q. 2135.00

KADAVI DAM
71.240
R.B.L. 568.700

M.W.L.602.750
F.R.L. 601.250
M.D.D.L 573.600
SPL.Q. 744.000

KASARI DAM
78.560
R.B.L. 581.060

M.W.L.623.300
F.R.L. 623.000
M.D.D.L 589.300
SPL.Q. 620.000

KUMBHI DAM
76.880
R.B.L. 572.120

M.W.L.612.700
F.R.L. 612.200
M.D.D.L 578.500
SPL.Q. 416.000

TULASI DAM
98.290
R.B.L. 570.720

M.W.L.617.910
F.R.L. 616.910
M.D.D.L 591.310
SPL.Q. 640.0

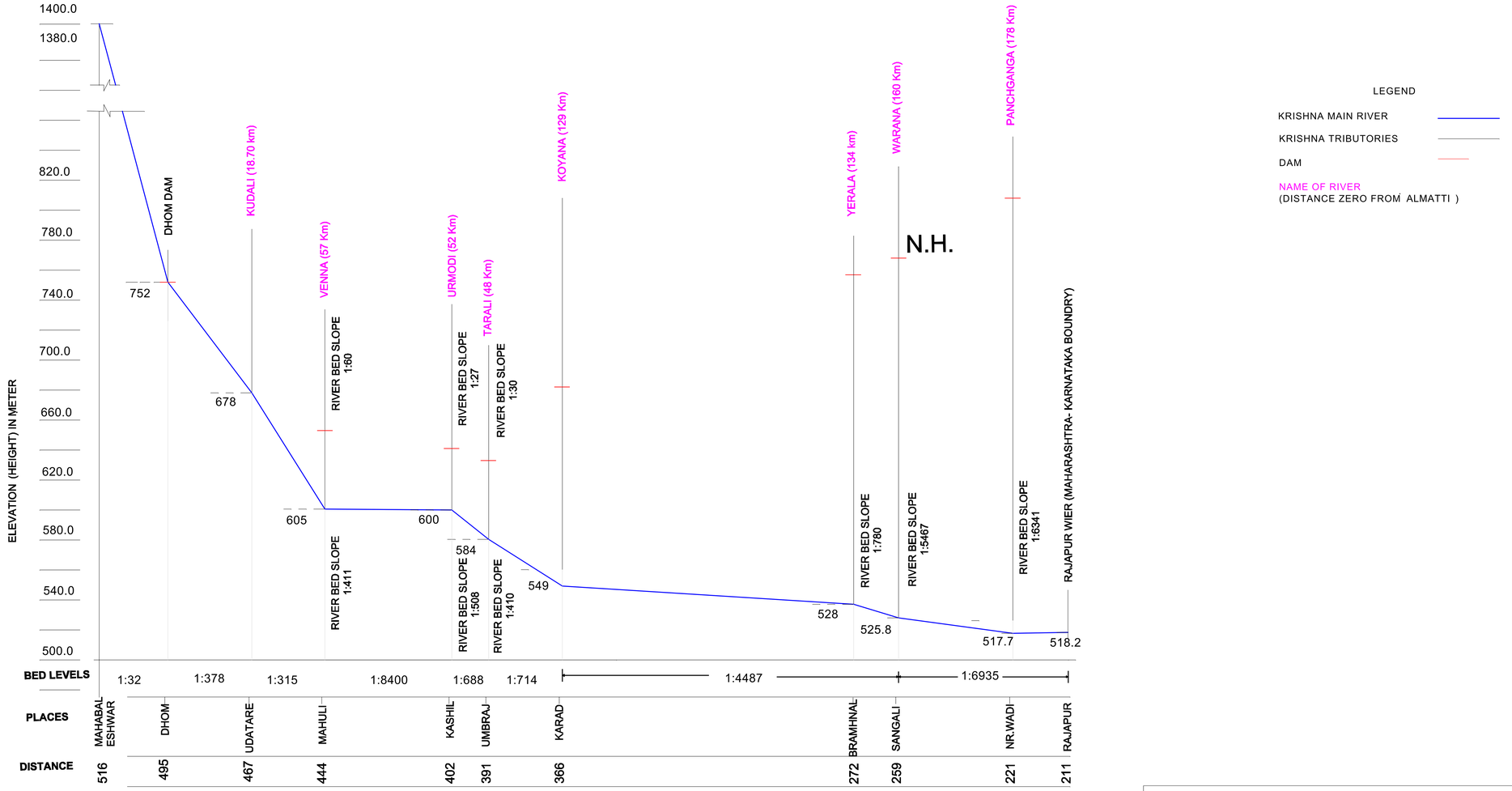
RADHANAGARI DAM
236.790
R.B.L. 553.820

M.W.L.591.280
F.R.L. 590.980
M.D.D.L 566.140
SPL.Q. 283.000

BHOGAWATI RIVER

NOTE :
DISTANCES NOT TO SCALE
CHAINAGE ZERO AT ALMATTI.

GOVT OF MAHARASHTRA
WATER RESOURCES DEPARTMENT
EXPERT STUDY COMMITTEE: FLOODS 2019
LINE DIAGRAM OF RIVER SYSTEM IN KRISHNA K-1 SUB-BASIN



L- SECTION OF KRISHNA RIVER AND TRIBUTORIES IN MAHARASHTRA

GOVT. OF MAHARASHTRA
WATER RESOURCE DEPARTMENT

EXPERT STUDY COMMITTEE : FLOODS 2019

L- SECTION OF KRISHNA RIVER AND TRIBUTORIES IN MAHARASHTRA

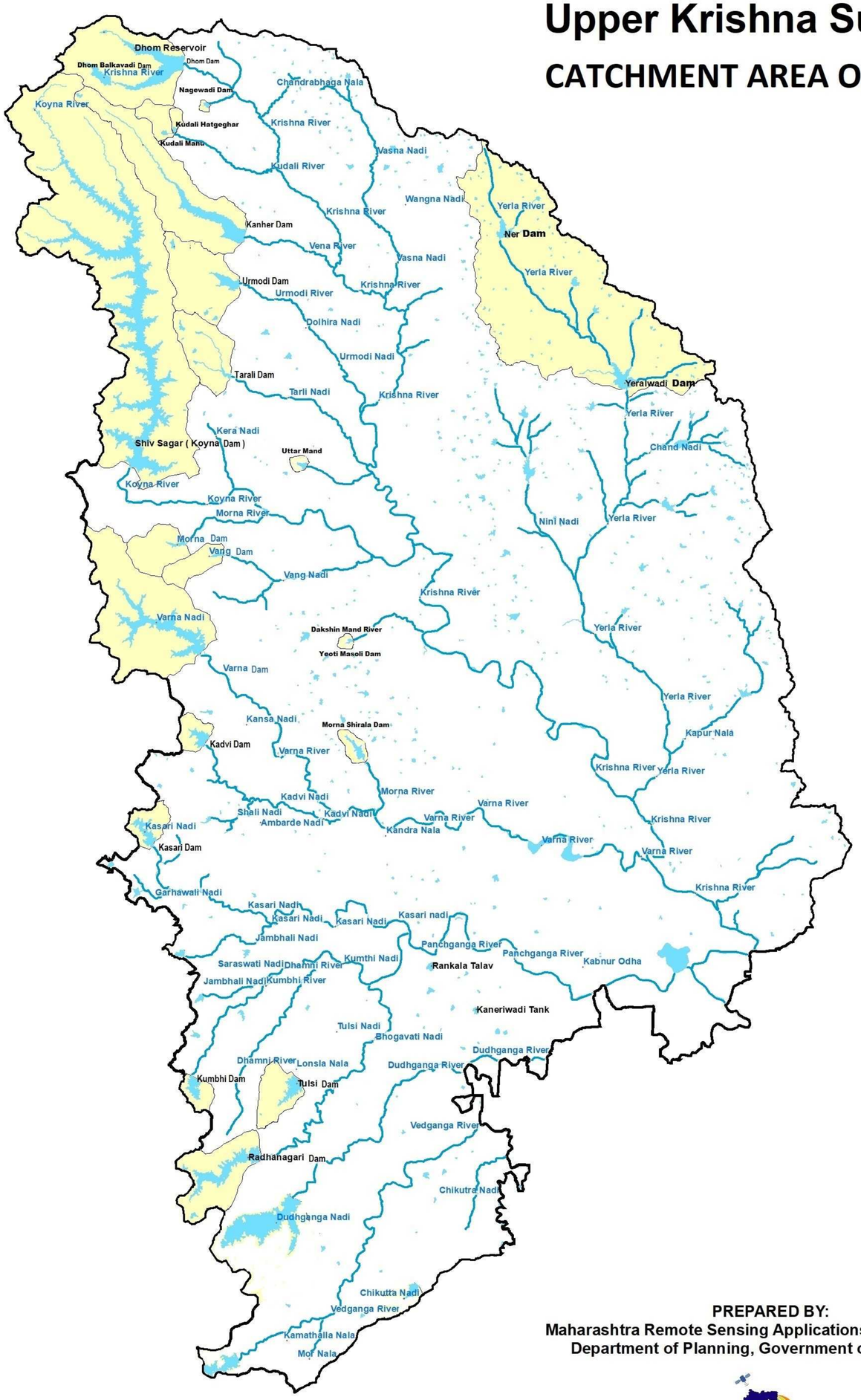
Upper Krishna Sub-Basin (K-1)

CATCHMENT AREA OF KRISHNA BASIN

0 12.5 25 50

Kilometers

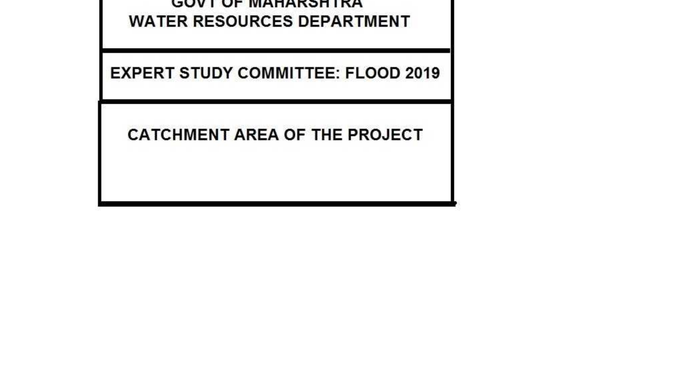
1:300,000



Legend

- River
- Dam Reservior
- Intercepted Catchment Area (3317.00 Sq.Km.)
- Free Catchment Area (13811.00 Sq.Km.)

Location Map of Krishna Basin - Maharashtra State



PREPARED BY:
Maharashtra Remote Sensing Applications Centre (MRSAC)
Department of Planning, Government of Maharashtra

MRSAC

GOVT OF MAHARSHTRA WATER RESOURCES DEPARTMENT
EXPERT STUDY COMMITTEE: FLOOD 2019
CATCHMENT AREA OF THE PROJECT

Krishna River Flood Vulnerability in Kolhapur District

Annexure - 2B

9

N
W E
S

0 3 6 12 18 24
Km

Shahuwadi
Panhala
Karvir
Hatkanangle
Shirol
Radhanagari
Bavda
Bhudargad
Ajra
Chandgad

District Index

Legend
State Boundary
District Boundary
Selected District

Taluka Index

Legend
Taluka Boundary
Selected District

Legend

- Observed Flood line 2019
- Railway
- National Highway
- State Highway
- Major District Road
- Other District Road
- Settlement
- State Boundary
- District Boundary
- Taluka Boundary
- Village Boundary
- Reservoirs
- Krishna Tributaries
- Inundation 09 Aug 2019
- Flood Vulnerability

Disclaimer:

The vulnerability thematic map is based on the various data obtained during flood 2019. The map shall be useful for taking preventive measures during flood event in future.

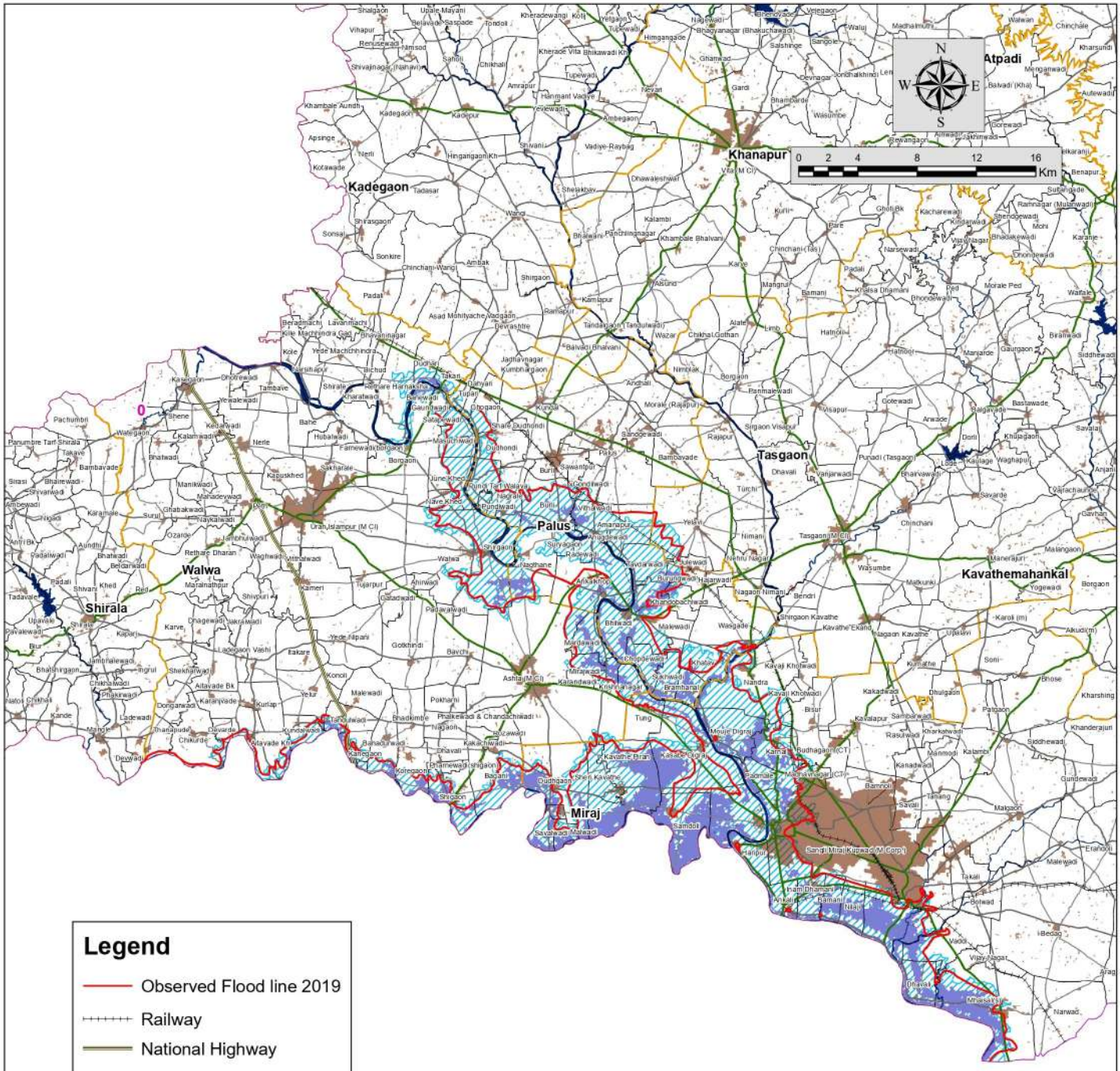
Data Analysis & Processing by

MAHARASHTRA REMOTE SENSING APPLICATION CENTRE

Autonomous Body of Department of Planning,
Government of Maharashtra. VNIT Campus,
South Ambazari Road, Nagpur – 440 010

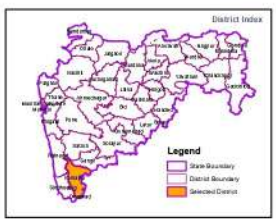
Krishna River Flood Vulnerability in Sangli District

Annexure - 2A



Legend

- Observed Flood line 2019
- Railway
- National Highway
- State Highway
- Major District Road
- Other District Road
- Settlement
- State Boundary
- District Boundary
- Taluka Boundary
- Village Boundary
- Reservoirs
- Krishna Tributaries
- Inundation 09 Aug 2019
- Flood Vulnerability



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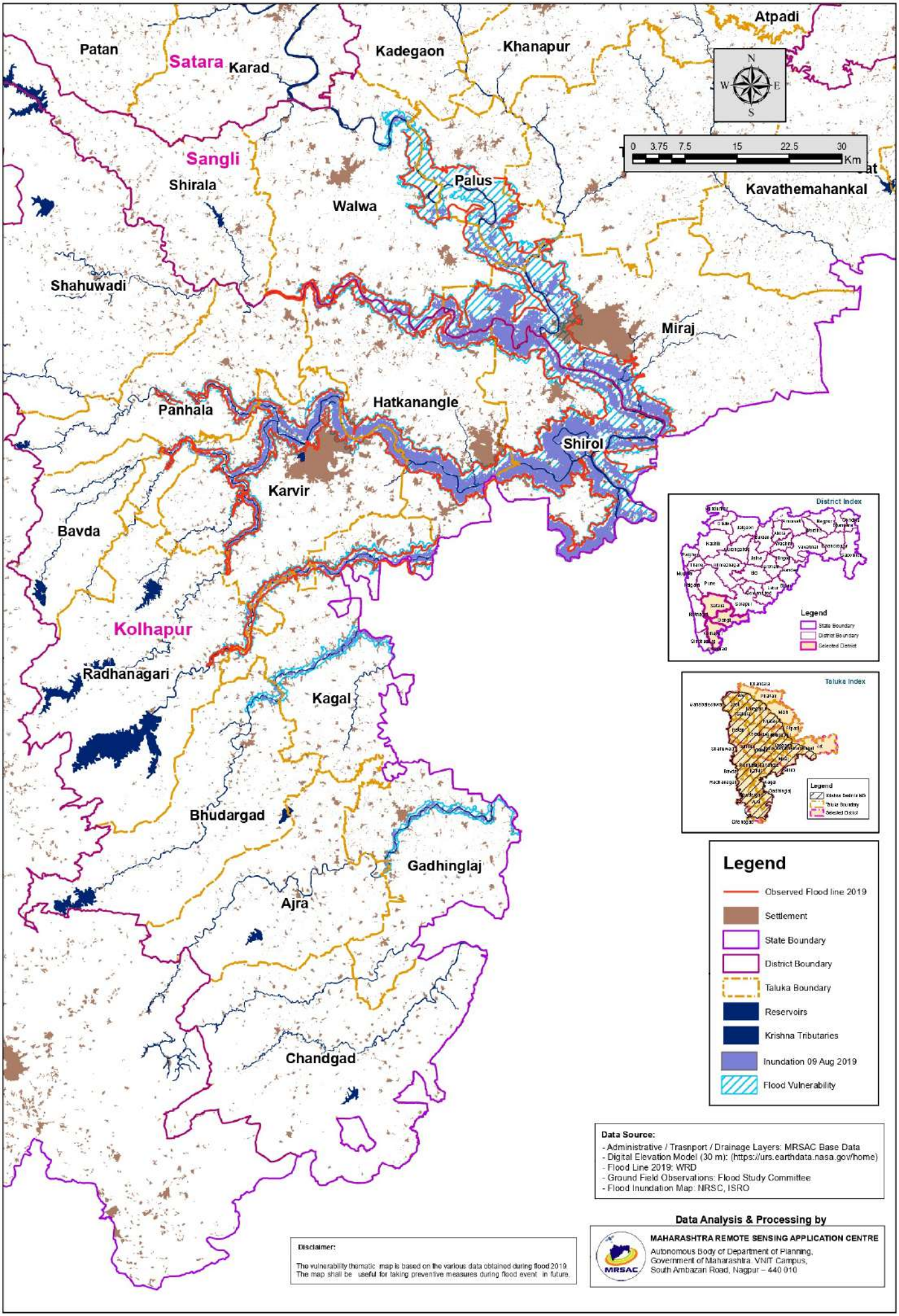
Krishna River Flood Vulnerability in Sangli & Kolhapur District

Annexure - 2



Data Source:

- Administrative / Trasnport / Drainage Layers: MRSAC Base Data
- Digital Elevation Model (30 m): (https://urs.earthdata.nasa.gov/home)
- Flood Line 2019: WRD
- Ground Field Observations: Flood Study Committee
- Flood Inundation Map: NRSC, ISRO

Disclaimer:

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